

Deepak Singh

(551) 376-8908 | singhdeepak2406@gmail.com | [linkedin.com/in/deepakSinghdev](https://www.linkedin.com/in/deepakSinghdev) | github.com/deepakSingh-dev | deepaksingh.me

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, C#, C, PHP, SQL, PL/SQL

AI & ML Engineering: LangChain, LangGraph, Hugging Face, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, SciPy, LSTM, RNN, Neural Networks, NLP, Prompt Engineering, Inference Pipelines, Agentic Workflows, RAG, Model Evaluation Methods, Linear Regression, Clustering, Random Forest, OpenMP

Backend & Frontend: FastAPI, Node.js, Next.js, React.js, Redux, GraphQL, REST APIs, Express, Electron, WebSockets, State Machine, Axios, Vite, esbuild, Radix UI, Angular

Data & Cloud: PostgreSQL, MongoDB, Redis, Kafka, Airflow, Spark, AWS (EC2, S3, Lambda, ECS, ElastiCache, CloudFormation), GCP, Docker, Kubernetes, Terraform, Ansible, Jenkins, Git, CI/CD, Apache Spark, Big Data, Hadoop, ETL, RabbitMQ, PGVector

DevOps & Tools: Jenkins, GitLab CI/CD, PyTest, Selenium, Grafana, Celery, MCP Server, Jira, Confluence, Agile/Scrum, SDLC, A/B Testing, MLOps, Microservices Architecture

EDUCATION

Stevens Institute of Technology Hoboken, USA
Master of Science (M.S.) in Information Systems (GPA: 3.9/4.0) Sep 2024 - May 2026
Relevant Coursework: AI for Software Engineering, Management of AI Technologies, Agile Methods for Software Development, Project Management, Data Analytics and Machine Learning, Big Data Management, Cyber Security Principles

Siddhartha Institute of Engineering and Technology Hyderabad, India
Bachelor of Technology (B.Tech.) in Computer Science Aug 2018 - Dec 2022
Relevant Coursework: Operating Systems, Data Structures and Algorithm, Artificial Intelligence, Computer Networks

PROJECTS

SpecAI - Constraints-First AI Coding Workflow | *Python 3.11, LangGraph, PydanticAI, FastAPI, PyTest, MCP Server, GitHub Actions, AWS ECS* Mar 2026

- Reduced untested AI-generated code by 70% in simulated team workflows by engineering a LangGraph state machine that enforces spec-first validation, requiring engineers to define inputs, outputs, and invariants before generation begins.
- Developed a PydanticAI-validated test generation pipeline converting natural language specs into typed PyTest cases, executed in isolated subprocesses via custom MCP server to block code generation until all assertions pass.
- Shipped GitHub Actions integration enforcing spec file presence on every pull request, automating quality gates and establishing a repeatable workflow adopted by 3 internal teams to reduce post-merge defects.

AgentGuard - AI Agent Observability & Reliability Platform | *Python, LangGraph, CrewAI, Redis, WebSockets, Prometheus, Grafana, React, MCP Server* Feb 2026

- Built production observability for LangGraph/CrewAI multi-agent systems with a drop-in Python SDK capturing every node transition, tool call, token cost, and latency with zero code changes to existing architectures.
- Designed real-time trace streaming via Redis pub/sub and WebSockets powering a React dashboard that renders live decision trees and cost attribution per feature, with Prometheus metrics enabling Grafana alerting on budget breaches.
- Implemented hallucination detection using Claude API scoring every LLM output for confidence, flagging low-quality responses before downstream propagation, reducing error cascade incidents by 65% in multi-agent simulations.

DocIntel - Enterprise Multi-Modal RAG System | *Python 3.11, LangGraph, PyMuPDF, Camelot, Claude Vision API, PGVector, Celery, Redis, Next.js, AWS S3* Jan 2026

- Reduced unsupported answers by 60% vs single-pass retrieval by engineering a LangGraph agentic RAG loop that self-assesses chunk quality and auto-re-queries with refined terms when initial confidence falls below threshold.
- Designed multi-modal ingestion pipeline handling PDFs, scanned images, and mixed-format documents by extracting text via PyMuPDF, tables via Camelot, and image captions via Claude Vision API before embedding into PGVector.
- Implemented PydanticAI-validated citation output mapping every answer to typed source references including file name and page number, cutting hallucination-related support tickets by 80% while async Celery processes uploads.

WORK EXPERIENCE

Software Development Engineer Oct 2023 - Jul 2024
RobotSpace Robotics and Automation Pvt Ltd Hyderabad, India

- Boosted operational efficiency 5x for 200+ facility users by architecting backend for ERP system's no-code dashboard module using Python, FastAPI, and PostgreSQL, enabling non-technical teams to customize production monitoring views via drag-and-drop and reducing support tickets by 40%.
- Cut production downtime by 30% through developing multi-platform factory monitoring suite (web, Android, HMI Linux devices) using high-volume Redis pipelines for real-time scheduling and anomaly detection, with WebSockets enabling instant alert propagation across 15+ production lines.
- Engineered navigation system and UI for autonomous material-transport robot using Python, Redis pub/sub, and shell scripting for real-time path planning across 50K+ sq. ft. facility, improving route efficiency by 22% and reducing manual intervention by 65%.
- Reduced release cycle time from 4 hours to 45 minutes by deploying containerized microservices via Docker with Jenkins CI/CD pipelines and Selenium test automation, improving deployment success rate by 15% and enabling daily production releases.

Associate Software Developer Jan 2023 - Sep 2023
GKB Labs LLP Hyderabad, India

- Increased sales team performance by 25% by developing custom CRM tool "Rep Donkey" using Laravel, PHP, jQuery, and PostgreSQL to automate 8 manual sales workflows, with optimized indexing accelerating data retrieval by 40%.
- Improved user engagement by 30% and cut operational costs by 20% by leading CMS database schema design and Agile/Scrum feature delivery across 3 cross-functional sprints, collaborating with product and QA teams to prioritize high-impact functionalities.
- Modeled predictive sales analytics database schemas supporting customer engagement insights, resulting in 40% faster query performance and enabling data-driven decision-making for sales leadership through automated reporting dashboards.

LEADERSHIP & ACTIVITIES

Stevens Graduate Technical Association Sep 2025 – May 2026
Vice President: Organized hackathons, technical workshops and industrialist speaker sessions. Hoboken, USA

Product Management Club Jan 2025 – May 2025
Volunteer: Coordinated 4+ club events per semester on product strategy, roadmap planning, and cross-functional collaboration for 50+ members. Hoboken, USA

Certifications

AWS Certified AI Practitioner (May 2026) | **Prompt Engineering Skills for Developers** - LinkedIn (Apr 2026) | **Delta Training for Automation** – Automation Anywhere (Dec 2024)